Nobody would ever claim that a bacterium was a conscious strategist, yet bacterial parasites are probably engaged in ceaseless games of Prisoner's Dilemma with their hosts and there is no reason why we should not attribute Axelrodian adjectives—forgiving, non-envious, and so on—to their strategies. Axelrod and Hamilton point out that normally harmless or beneficial bacteria can turn nasty, even causing lethal sepsis, in a person who is injured. A doctor might say that the person's 'natural resistance' is lowered by the injury. But perhaps the real reason is to do with games of Prisoner's Dilemma. Do the bacteria, perhaps, have something to gain, but usually keep themselves in check? In the game between human and bacteria, the 'shadow of the future' is normally long since a typical human can be expected to live for years from any given starting-point. A seriously wounded human, on the other hand, may present a potentially much shorter shadow of the future to his bacterial guests. The 'temptation to defect' correspondingly starts to look like a more attractive option than the 'reward for mutual cooperation'. Needless to say, there is no suggestion that the bacteria work all this out in their nasty little heads! Selection on generations of bacteria has presumably built into them an unconscious rule of thumb which works by purely biochemical means.

Plants, according to Axelrod and Hamilton, may even take revenge, again obviously unconsciously. Fig trees and fig wasps share an intimate cooperative relationship. The fig that you eat is not really a fruit. There is a tiny hole at the end, and if you go into this hole (you'd have to be as small as a fig wasp to do so, and they are minute: thankfully too small to notice when you eat a fig), you find hundreds of tiny flowers lining the walls. The fig is a dark indoor hothouse for flowers, an indoor pollination chamber. And the only agents that can do the pollinating are fig wasps. The tree, then, benefits from harbouring the wasps. But what is in it for the wasps? They lay their eggs in some of the tiny flowers, which the larvae then eat. They pollinate other flowers within the same fig. 'Defecting', for a wasp, would mean laying eggs in too many of the flowers in a fig and pollinating too few of them. But how could a fig tree 'retaliate'? According to Axelrod and Hamilton, 'It turns out in many cases that if a fig wasp entering a young fig does not pollinate enough flowers for seeds and instead lays eggs in almost all, the tree cuts off the developing fig at an early stage. All progeny of the wasp then perish.'